Ran Tao

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Full-Stack (iOS + React) Software Developer. Student Researcher in AI for Science.

Launched an iOS app that garnered 1,900+ downloads across 40+ countries. CTO and co-founder of a start-up company. The company was accepted into the X-Lab incubator at Tsinghua University (#1 ranked incubator in China).

EDUCATION

University of Southern California *Sep 2025 - June 2027 (Expected)*
Master of Science in Computer Science| GPA: 3.8/4.0 Los Angeles, CA

University of California – Los Angeles *Sep 2022 - Mar 2025*
Master of Science in Chemistry | GPA: 3.8/4.0 Los Angeles, CA

New York University *Sep 2018 - Jan 2022*
Bachelor of Science in Chemistry, Minor in Physics | GPA 3.7/4.0 New York City, NY

WORK EXPERIENCE

ByteDance Inc. *May 2026 – Current*
Global Payment – Software Engineer Intern (Frontend) San Jose, USA

- **Individually shipped** end-to-end event tracking (page register, click, and jump) for TikTok Buy Now Pay Later (Southeast Asia). Built bill/repayment flows, enabling user path visibility for **~2M daily users**. Extended coverage through a three-phase production rollout (0% → 25.2% → 100%), supporting conversion funnel drill-down.
- **Web Development**: Developed business-context tracking instrumentation. Built shared payload builders and container passthrough so institution/region/viewtime fields travel stably from native/H5 into the WebView tracking chain.
- **System Design**: Developed and migrated the infrastructure of KYC (a regulatory compliance system for payment services) in TikTok Creator Marketplace. Identified a session auth error caused by cookie overwrite under shared internal payment-domain cookies, and fixed this error by shifting to header auth. Improving KYC session reliability for in-app WebView traffic.
- **User Experience**: Redesigned file upload to stream via base64-in-payload to async upload with file_id references, fixing silent ID-photo drops and reducing submit body size by 90%+, improving features for better user experience.

Beijing Landoo Intelligence Travel Technology Co., Ltd *Dec 2024 – June 2025*
Software Engineer (Full-Stack – iOS App Development) Remote, USA

- **Founder and developer** of *Visit China: All in 1 @ChinaGo* (link), an iOS app designed to empower international travelers to explore China with ease and confidence. It allows translation to more than 40+ languages from Chinese in multi-modal scenarios, including real-time speech translation, translation based on images and offline translation. It also enables smart navigation and llm-agent based auto-planning functionality based on user requests. It has garnered **1,900+ downloads** from **40+ countries** and was accepted at the ***X-Lab incubator at Tsinghua University***.
- **iOS Development**: Developed a fully functional iOS app using **Swift/SwiftUI**, optimized image loading & state management, improving app average load time by 35%, implemented automated user event tracking (100% of feature clicks & sessions).
- **Backend and Database**: Built a **serverless backend** using Supabase Edge Functions and persisted data through **PostgreSQL**, enabling elastic scaling based on API traffic, allowing 1M+ API requests/sec with <200ms latency.
- **Data Analysis**: Built comprehensive online and offline user behavior logging using Python (Scikit-learn, PyTorch, Matplotlib, Pandas, etc.), with waterfall stages and real-time analysis, providing strong direction hints for user preferences and demands.
- **Business side**: Planned the company's roadmaps based on business analytics data, led marketing decisions to discover new users.

Neurova *Aug 2025 – Dec 2025*
Software Engineer (FrontEnd – AI/ML Platform) Intern Remote, USA

- Developed a Flask-based app serving real time S&P 500 stock sentiment analysis in for investors to make investing decisions.
- Implemented the user authentication system to manage user sessions, with a log-in/log-off function and password hashing using Flask-login. Utilized ploty to present visualization of sentiment analysis in time-series fashion in an interactive fashion.
- Improved backend scraping efficiency of by 100X by switching from real-time to batched fashion, gathered numerous online finance resources for key-words capturing, aggregated and cleaned data for VADER Positive/Negative sentiment scoring.
- Finetuned and served DeepSeek using Ollama and Financial PhraseBank dataset, improve sentiment analysis accuracy by 30%.

Shanghai AI Lab *June 2025 – Aug 2025*
Machine Learning Engineer Intern Shanghai, China

- Built a **Google Colab online platform** for versatile protein structure computing and fast retrieval of models for chemists.
- Clustering a 200 M protein dataset as an encoder to a new Boltz-2 or Protonix head, achieving 30% faster inference.
- Developed Diffusion based models (Distributional Graphormer, DiG model) with **PyTorch** for large-scale protein conformation prediction, integrating fast MSA sampling and clustering, with customized PPFT loss and configurable positional biases.
- Managed an elastic and distributed GPU clustering on Aliyun to smoothly train models, reducing training cycles by 31%.

RESEARCH EXPERIENCE

UCLA Doyle Lab of Chemistry | AbbVie *Sep 2022 – July 2024*
Chemistry Machine Learning Researcher Los Angeles, CA

- Leading collaboration with AbbVie. Automated high-throughput screening workflows, built reproducible data pipelines (**Python**, **Scikit-learn**, **PyTorch, UMAP/PCA clustering**) and managed HPC (Slurm) clusters to accelerate scalable drug synthesis.
- Developed a machine learning–driven small molecule synthesis pipeline, integrating **Bayesian optimization, Graph Neural Network**, and **Multivariate Analysis** to predict reaction yields utilizing **Python** with 0.85+ R^2, reducing lab workload by 40%.
- Co-authored a ***peer-reviewed paper*** published in the *European Journal of Organic Chemistry* (2024), demonstrating data-driven design for Ni-catalyzed Suzuki–Miyaura couplings.

UCLA Shafaat Lab of Chemistry *Sep 2022 – July 2024*
Chemistry Data Science Researcher Los Angeles, CA

- Engineered human butyrylcholinesterase (BChE) via Rosetta/ProteinMPNN by inserting a histidine-rich metal-binding site, to achieve enhanced catalytic efficiency in organophosphate (OP) neurotoxin degradation (potential chemical warfare antidote applications).
- Automated enzyme design pipelines by integrating Python-XML scripting with AlphaFold3 structural prediction, enabling high-throughput mutagenesis screening on High-Performance Computing cluster (10-fold faster than previous workflows).
- Validated mutant libraries through transition-state docking and saturation mutagenesis, identifying 4 variants with sub-micromolar binding affinity to OP transition-state analogs

Undergraduate Research Assistant | Professor Shaolin Zhu *Sep 2022 – July 2024*
Nanjing University Nanjing, CN

- Preliminary experiments about NiH-catalyzed asymmetric hydroamination using nitrobenzene as aminating reagent.
- Optimized the reaction conditions of Ni-H catalyzed hydroalkynylation of α,β-unsaturated amides with bromoalkyne using different chiral ligands

Publications

1. LeSueur A, **Tao R**, Doyle A, Sigman M. Multi-Threshold Analysis for Chemical Space Mapping of Ni-Catalyzed Suzuki-Miyaura Couplings. *EJOC* **2024**, *27*, e202400428, DOI: 10.1002/ejoc.202400428

   Developed a framework to prioritize ligands for drug-scale cross-coupling reactions.

2. He Y, **Tao R**, Zhu S. Nickel-Hydride Catalyzed Remote Hydroarylation of Olefins. *Synlett* **2021**, *33*, 224-230, DOI: 10.1055/a-1637-9308

3. Cheng Y, Gui Z, **Tao R**, Wang Y, Zhu S. NiH-Catalyzed Asymmetric Hydroalkynylation of α,β-Unsaturated Amides, *Green Synthesis and Catalysis* **2021**, *3*, 377-379, DOI: 10.1016/j.gresc.2022.03.009

4. Song T, Li T, **Tao R**, et al., CuO/g-C_3N_4heterojunction photocathode enhances the microbial electrosynthesis of acetate through CO_2 reduction, *Science of the Total Environment* **2021**, *818*, 151820, DOI: 10.1016/j.scitotenv.2021.151820

5. Raab T J, Ho T C, Kerr T, **Tao R**, Shekhar S, Swiatowiec R, Haibach M, Doyle A. DiploPhos: A Hemilabile Bisphosphine for Sterically Hindered Ni-Catalyzed Suzuki–Miyaura Couplings. *ChemRxiv* **2025**, DOI: 10.26434/chemrxiv-2025-q7td5

SKILLS

Programming and Tools:	Python, Java, C++, C, Swift, JavaScript, TypeScript, SQL, Flask, React, Node.js, Supabase, PostgreSQL, MySQL, MongoDB, Redis, Docker, Kubernetes, Git
Machine Learning:	Generative AI Deployment, TensorFlow, Pytorch, Scikit-Learn, Finetuning
Cloud Computing:	HPC (Slurm), AWS (S3, EC2, Lambda), Google Cloud (BigQuery, Vertex AI), Aliyun